

RateController User Manual

RateController (RC) is a Windows application for controlling the application rate of agricultural products (liquids, granules, seeds, ammonia) in conjunction with AgOpenGPS. It communicates with rate control modules via UDP/Ethernet to manage product application rates, boom sections, and tank quantities.

RC can also be used without AgOpenGPS. Automatic rate control based on ground speed will continue to work normally. Section control will be manual only.

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1. Getting Started

Before operating, complete these setup steps in order:

1. **Create or load a Profile** — a profile stores all machine settings (products, sections, relays, etc.)
2. **Connect the module** via Ethernet — go to **Modules > Communication**, set the subnet, and send it to the module
3. **Load a board preset** — go to **Modules > Boards** and select the Example Preset matching your hardware
4. **Configure the module** — go to **Modules > Config** to set the Module ID, description, and relay type
5. **Configure pins** — go to **Modules > Pins** to verify or set sensor and PWM pin assignments
6. **Configure relay pins** — go to **Modules > Relay Pins** to assign physical pins to each relay
7. **Set valve mode** — go to **Modules > Valves** to select 2-wire or 3-wire valve control
8. **Define sections** — go to **Machine > Sections** and set section widths
9. **Configure relays** — go to **Machine > Relays** and assign relay types to sections
10. **Configure switches** — go to **Machine > Switches** to set master switch behavior
11. **Set up your product** — go to **Products > Rate** and enter product name, control type, units, and rates
12. **Assign the sensor** — go to **Products > Settings** to assign the module ID and sensor ID
13. **Calibrate** — go to **Machine > Calibrate** to determine the Cal Factor
14. **Create a Job** — go to **File > Jobs** to create a job for the field
15. **Connect AgOpenGPS** — the AOG status indicator should turn green

Data Folder (File > Folders)

RC stores its data — profiles, jobs, fields, rate maps, and logs — in a data folder (Documents by default). Use **File > Folders** to move it, e.g. to a synced folder (OneDrive, Dropbox) or another drive.

Control	Description
Current location	The data folder currently in use
New location	Press to browse for the new data folder; the chosen path can also be typed in the box
Copy Data	Copy the contents of the current data folder to the new location. Leave unchecked to simply switch to a folder that already contains RC data.
Overwrite	Allow copying into a folder that already contains files. Without this, Copy Data only copies into an empty folder.

Press **OK** and confirm — RC changes the folder and restarts. The new location cannot be a sub-folder of the current data folder.

2. Profiles

Menu: File > Profiles

A profile stores all machine configuration settings. Multiple profiles can be maintained for different implements.

Button	Action
 1. New	Create a new profile
 2. Load	Load the selected profile
 3. Delete	Delete the selected profile
 4. Copy	Copy the selected profile to a new name
5. File name	Enter name for New or Copy
6. Files	List of available profiles to select
  7. Page Up/Down	Scroll through the profile list

3. Jobs

Menu: File > Jobs

Jobs track application activity per field. Each job stores application data, notes, and a rate map.

Current Job

Field	Description
Name	Job name
Date	Date/time of the job
Field	Select from list or type a new field name
Notes	Free-text notes for the job

Jobs List

The list shows each job's **Name**, **Date**, and **File**. Click a job to select it.

Control	Action
Year	Year used by the filter
Field	Field used by the filter
Filter	When checked, only jobs matching the selected Year and Field are listed
 Field	Delete the selected field from the Fields list (not allowed while the field is in use)
 Load	Open the selected job
 New	Create a new job
 Copy	Copy selected job and its rate map to a new file
 Delete	Delete selected job
Show at Start-Up	Display the Jobs menu when the app launches
 Activity	Job report
 Import	Import jobs
 Export	Export all jobs to a chosen folder

4. Options – Speed Source & Display

Menu: File > Options

Selects the source of ground speed used for rate calculations and configures display units.

Speed Source

Option	Description
GPS	Use speed from AgOpenGPS
Simulated Speed	Simulate a constant speed (useful for testing)
Wheel Sensor	Use a speed sensor connected to a module GPIO pin

Wheel Sensor Setup

- 1. Enter the **Module ID** and **GPIO pin** used for speed sensing.
- 2. Enter the calibration number in **pulses per mile** or **pulses per kilometer**.
- 3. Press the **up arrow** to send configuration to the module (also sent on Save).
- 4. Use the **Calibration screen** to calculate the calibration number if unknown.

Display Options

Option	Description
Metric	Toggle between metric (km, ha, L) and imperial (mi, ac, gal) units
Rate Display	Show or hide an additional rate display window with a larger font

5. Products – Rate

Menu: Products > Rate

Configures the fundamental properties of a product.

Setting	Description
1. Product Name	Name of the product
2. Control Type	How the application rate is controlled (see below)
3. Quantity Units	Units used to measure the product (L, gal, kg, lbs, seeds, etc.)
4. Coverage Units	Acre, Hectare, Minute, or Hour
5. Sensor Counts/Unit	Number of sensor pulses per unit of product applied
6. Density	Product density (used for mass/volume conversion)
7. Base Rate	Target application rate
8. Alt. Rate (%)	Alternate rate as a percentage of the base rate
9. Tank Size	Total tank capacity

Control Types

Selects the actuator driving the product and how the PID loop positions it. This must match the physical valve/motor wiring on the module — an incorrect selection can leave the actuator open or unresponsive.

Type	Description
Standard Valve	A proportional flow-control valve driven continuously by the PID loop — includes a motorized ball valve or a solenoid valve. PWM = 0 stops the valve motor and holds its last position (does not drive it closed) — use this for valves with no dedicated closed/shutoff position.
Combo Close	Same continuous PID control as Standard Valve — includes a motorized ball valve or a solenoid valve — but the valve is also driven fully closed whenever the product/section is off (rather than just holding position). Use for valves that must shut off flow, not just stop adjusting.
Motor	Drives a variable-speed motor that IS the metering device itself (e.g. a hydraulic or electric drive motor turning a metering roller/auger/pump) — not a motorized valve. PID output accumulates over time (integrating control) instead of positioning directly, so it needs different gain tuning than a valve.
Combo Timed	For close-only/on-off style valves that can't be positioned proportionally. Instead of continuous PWM, the valve is pulsed through alternating Adjust/Pause cycles (set on the Control page) to work the rate toward target, with a fast minimum-start ramp from zero.

6. Products – Settings

Menu: Products > Settings

Configures sensor assignment, alarm thresholds, and display preferences per product.

Setting	Description
1. Enable product	Activate this product
2. Rate Sensor Location	Module ID (0–7) and Sensor ID (0–1) for the flow sensor
3. Minimum UPM	Minimum flow rate the controller will maintain. Enter either a fixed UPM value or a By Speed value (MPH/KPH) — the hint below the boxes shows the equivalent value the other way (e.g. the speed that corresponds to the entered UPM).
4. Off-Rate Alarm	% error in rate that triggers an alarm
5. Default Product	Product displayed at startup
6. Scale Weight	Show the scale weight screen

Controls

Control	Action
 Navigate left	Previous product
 Navigate right	Next product
 Cancel	Discard changes
 Save	Save settings

7. Products – Mode

Menu: Products > Mode

Sets the operating mode for each product. Navigate between products with the left/right arrows.

Mode	Description
Controlled UPM	PID feedback control — adjusts motor/valve to maintain the target rate as speed changes
Constant UPM	Maintains a fixed flow rate based on total machine width regardless of which sections are on. Used for sprayers where turned-off sections return flow to the tank (after the flow meter). RC corrects the measured flow by subtracting the estimated contribution of the off sections to give an accurate application rate per acre/hectare.
Document Applied	Tracks and records material that has already been applied — no active control
Document Target	Tracks and records the target rate for comparison — no active control

Controls

Control	Action
 Navigate left	Previous product
 Navigate right	Next product
 Cancel	Discard changes
 Save	Save settings

8. Products – Data Reset

Menu: Products > Data

Displays accumulated totals for the selected product and allows resetting individual counters. Navigate between products with the left/right arrows.

Counter	Description
Coverage Area 1	Total area covered (primary counter)
Coverage Area 2	Total area covered (secondary counter)
Quantity Applied 1	Total product applied (primary counter)
Quantity Applied 2	Total product applied (secondary counter)
Hours 1	Operating hours (primary counter)
Hours 2	Operating hours (secondary counter)

Press the reset button next to any counter to zero it. Both counters are independent, allowing a session total and a season total to be tracked simultaneously.

Controls

Control	Action
 Navigate left	Previous product
 Navigate right	Next product
 Reset counter	Zero the selected counter
 Cancel	Discard changes
 Save	Save settings

9. Products – Monitoring

Menu: Products > Monitoring

Displays real-time diagnostic data for the selected product. Navigate between products with the left/right arrows.

Rate Information

Display	Description
Target UPM	The current setpoint (desired flow rate)
Applied UPM	The measured actual flow rate
Rate Error %	Difference between target and actual, with sign
PWM	Current PWM output value (0–255)
Frequency (Hz)	Flow sensor pulse frequency

Speed & Coverage Information

Display	Description
Ground Speed	Current speed in MPH or KPH
Working Width	Implement working width in feet or meters
Work Rate	Area covered per hour (ac/hr or ha/hr)

Section Status

Shows 16 section indicator boxes — green for On, red for Off.

Module Communication

Display	Description
WiFi / Signal	Module signal strength (RSSI) bar
Counts per Revolution	The number of pulses the sensor produces per revolution. Total pulses per minute are divided by this value to calculate RPM. For example, a value of 3 means every 3 pulses equals one revolution.

Controls

Control	Action
 Navigate left	Previous product
 Navigate right	Next product
 Cancel	Discard changes
 Save	Save settings

10. Products – Control (PID Tuning)

Menu: Products > Control

Configures the PID controller that adjusts the motor or valve to maintain the target rate.

Primary Settings

Setting	Description
1. Gain	Rate adjustment factor — how strongly the controller responds to rate errors. The effect is linear: doubling Gain doubles the correction. Errors are measured as a percentage of the target rate, so the same Gain setting behaves consistently at low and high rates.
2. Integral	Trims residual error once the rate is close to target (inside the BrakePoint band — see below). Far from target, only Gain (proportional) drives the approach; Integral does not speed that up. Increase if the rate settles near, but not exactly on, target. Integral accumulates once per PID loop tick, so a shorter PID loop time (setting 4 below) also makes it trim faster in real time, independent of this setting.
3. Max Power	Maximum PWM power delivered to motor or valve
4. Min Power	Minimum PWM power delivered to motor or valve

Adjustment Process

1. **Initial Setup** — Set minimum power to the minimum needed to move the motor/valve. Set maximum power to 100. Set Integral to 0.
2. **Gain Adjustment** — Slowly increase gain until the system overshoots the target, then reduce slightly to stabilize.
3. **Integral** — If the rate approaches target but settles slightly off, increase Integral to close the remaining gap. It only acts once you're already close (see BrakePoint) — it will not speed up a slow ramp toward target.

Optional / Advanced Settings

Settings 7–9 apply to the Combo Timed control type only; setting 5 applies to Motor only.

Setting	Description
1. Deadband	Error % below which no adjustment is made. Also controls when accumulated Integral is reset — only once the error clearly crosses to the other side of target, not on small wobble near the setpoint.
2. BrakePoint	Error % (of peak target) defining the "near target" band. Outside this band, only fast proportional (Gain) control acts and Integral stays at zero. Inside it, proportional gain drops to the Slow Adjustment % and Integral begins accumulating.
3. Slow Adjustment	% of full proportional adjustment used once inside the BrakePoint band
4. PID loop time	How often (50–250 ms) the PID recalculates its output. A shorter time makes the valve/motor react sooner to a real rate change, and also speeds up Integral trimming in real time (see Integral, above) since it accumulates once per tick. Reducing this after Gain/Integral were tuned at a slower rate can introduce new oscillation — check for overshoot as with any Gain increase.
5. Slew Rate	Motor control type only — maximum total PWM change per PID loop. Has no effect on Standard/Combo Close valves, which are positioned fresh each loop rather than ramped.
6. Max Integral change	Maximum Integral PWM change per PID loop. The total accumulated Integral is separately capped to the Max Power – Min Power range, so it can reach full authority (e.g. to free a stuck valve) without jumping there in one loop.
7. Min Start %	Combo Timed control type only — % of target rate below which the adjust/pause cycle is skipped for a faster start from 0.
8. Adjust time	Combo Timed control type only — milliseconds the valve is actively adjusted during each cycle.
9. Pause time	Combo Timed control type only — milliseconds the valve pauses (holds) between adjustments.
10. Min flow sensor Hz	Pulse frequency below which the sensor is treated as stopped/no-flow
11. Max flow sensor Hz	Pulse frequency above which pulses are rejected as sensor noise/bounce
12. Flow sample time (Smp Tm)	Time window in milliseconds (50–1500) used to calculate the median flow reading. A longer window smooths the reading; a shorter window responds faster. Default 400 ms.

Controls

Control	Action
 Navigate left	Previous product
 Navigate right	Next product
 Load defaults	Restore default PID values
 Open rate graph	Display the rate graph
 Gain/Integral/Power +	Increase the selected value
 Gain/Integral/Power -	Decrease the selected value
 Show advanced settings	Expand the advanced/optional settings panel
 Cancel	Discard changes
 Save	Save settings

11. Machine – Calibrate

Menu: Machine > Calibrate

Use this screen to determine the **Cal Factor** (pulses per unit) for a product. The goal is to achieve a consistent meter roller RPM or liquid flow rate matching typical field operation.

Step-by-Step Calibration

① Step 1 — Set Ground Speed

- Enter the ground speed you typically operate at.

② Step 2 — Set Meter Roller RPM / Valve Position

- Switch the product **ON** using the power button.
- Enter an **Initial Base Rate** and an **Estimated Cal Factor**.
- Press **Start** — RC adjusts flow to match the target rate.
- If successful, RPM is locked.
- The PWM value used is displayed under the lock (for motor controllers).
- RC stops automatically when the target rate is met.

If RC fails to reach target RPM:

- Press **Stop**.
- Check ground speed.
- Adjust the estimated Cal Factor, PWM min/max, or metering drive range.

③ Step 3 — Set Actual Cal Factor

- Press **Start** to run the meter roller and collect a sample.
- Press **Stop**, then enter the measured amount collected.
- RC calculates and displays the new Cal Factor.
- Press **Save** to record it.

④ Step 4 — Optional Adjustments

- To redo meter speed: press **Unlock**, then repeat Step 2.
- To reuse a previously saved setting: press **Lock**, then proceed to Step 3.

Important Notes

- AgOpenGPS is **not** used during calibration.
- The Switchbox is **not** used during calibration.
- If a switchbox is present, ensure the **master switch is OFF** before calibrating.
- Calibration uses **Constant UPM, Mode 2**.
- All sections on the calibrating module are activated during calibration; modules that are not calibrating are unaffected.

- More than one product can be calibrated at the same time, on the same or different modules (requires up-to-date module firmware).
- After calibration, the original mode and section states are restored.

Controls

Control	Action
 Start	Start the meter roller / begin collecting a sample
 Stop	Stop the meter roller
 Lock/Unlock	Lock or unlock the current RPM setting
 Power on/off	Switch the product on or off
 Cancel	Discard changes
 Save	Save the calculated Cal Factor

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12. Machine – Sections

Menu: Machine > Sections

Defines the boom sections and their widths.

- There are **16 sections per module** and up to **8 modules**.
- Module 0 = sections 1–16
- Module 1 = sections 17–32, etc.
- Sections can be defined **individually** or grouped into **zones**.
- All sections in a zone share the same width.
- All sections in a zone are controlled by one switch.
- AgOpenGPS can still auto-control each section individually within a zone.
- When creating a new profile, sections must be defined.
- Sections are also automatically redefined when AgOpenGPS section count changes.

Control	Action
 Copy Button	Copies the width from Section 1 to all other sections
 Cancel	Discard changes
 Save	Save settings

13. Machine – Relays

Menu: Machine > Relays

Assigns relay functions to each relay output on the module. There are **16 relays per module** and up to **8 modules**.

Control	Action
 Reset (circle)	Reset all relays to default
 Renummer (+)	Renummer sections sequentially
Relay Number	The relay output number
Relay Type	The behavior of the relay (see below)
Number	For Switch type: switch number. For Section/Invert_Section type: section number.
 Cancel	Discard changes
 Save	Save settings

Application Control

Spraying occurs when:

- Master is ON
- A section is enabled
- In Auto, the guidance system allows application

Relay Types

Type	Description
Section	Controls an individual section. In Manual, the switch turns the section on or off. In Auto, the guidance system controls the section, but the switch can disable it.
FlowMaster	Controls overall product flow. In Manual, it follows the Master switch. In Auto, it is on only when the Master switch is on and the machine is moving. Use this to control overall product flow through an upstream valve or common shutoff.
Invert_FlowMaster	Opposite of the FlowMaster output — on when the master switch is off or (in Auto) the machine is stopped. Intended for valve wiring that requires an inverted signal, not as a bypass for section activity. Use Bypass if you need a relay that responds to whether sections are actually on.
Bypass	On when all sections are off; off when any section is on. Use this for a return or bypass valve that diverts product when no boom sections are applying.
Master	Follows the Master switch. Use this when a relay should follow the master logic itself rather than actual flow-enabled status.
Invert_Master	Opposite of the Master output (on when the Master output is off).
Slave	On when any section is on; off when all sections are off.
Invert_Section	Opposite of the Section output (on when the section is off).
Power	Always on.
Switch	Controlled directly by a switch (not affected by Master or Auto).
Hyd Up	Hydraulic raise.
Hyd Down	Hydraulic lower.
Tram Right	Tram marker right.
Tram Left	Tram marker left.
Geo Stop	Geofence stop output.
None	No function assigned to the relay.

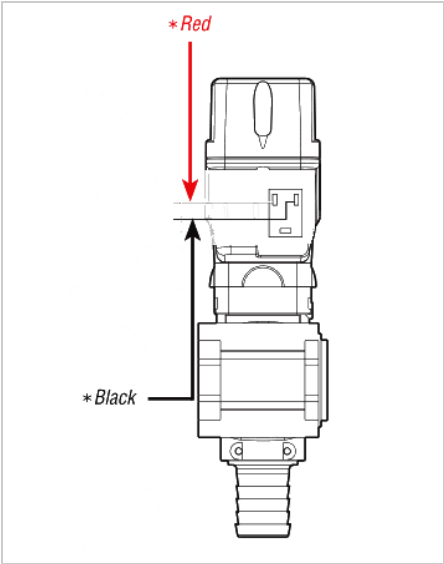
FlowMaster Valve Options

These options apply only when exactly one **FlowMaster** relay is defined on the selected module.

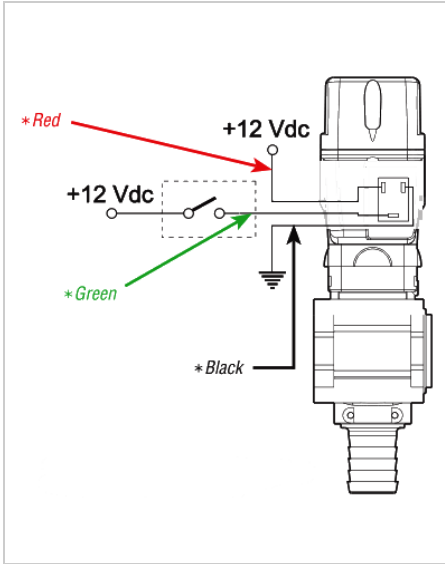
Option	Description
2-wire valve (single relay powers open and close)	One FlowMaster relay drives both open and close directions using the module's H-bridge output. Requires module support (e.g. RC15).
2-wire valve (2 relays: FlowMaster powers open, Invert_FlowMaster powers close)	Two relay outputs: the FlowMaster relay supplies power to open the valve; the Invert_FlowMaster relay supplies power to close it. Use this on modules without H-bridge support (e.g. RC11-2).
3-wire valve (relay signal, valve self-powered)	One FlowMaster relay provides a signal. The valve uses its own power supply to open or close based on that signal.

Valve Wiring Reference

The diagrams below show the wiring difference between 2-wire and 3-wire valves.



2-wire valve



3-wire valve

14. Machine – Switches

Menu: Machine > Switches

Configures how the master switch, auto switches, and on-screen virtual switches behave.

Master Switch Mode

Select one option. Controls what the master switch turns on and off.

Option	Description
Standard	Master switch controls the master relay and all section relays. When master turns off, all section relays turn off. When master turns back on, section relays resume their switch positions.
Master Override	Forces the module to treat master as always on, regardless of switch state. Useful for motor control when the master switch should not interrupt the motor PID loop.

Switch Type

Select one option. Describes the physical type of master switch connected.

Option	Description
Momentary	Standard push-to-make switch. Each press toggles master on or off. Holding in the On position activates Primed Start after the configured delay.
Maintained	Toggle or latching switch. Master follows the switch position directly — on when switch is on, off when switch is off. Primed Start is not available in this mode.

Other Settings

Setting	Description
Work Switch	Uses a switch connected to the implement to gate master on/off. The switch's pin, momentary behavior and NO/NC invert are set per module under Modules > Pins.
Auto Switch – Rate	Enables automatic rate control via the switchbox Auto Rate switch
Auto Switch – Sections	Enables automatic section control via the switchbox Auto Section switch
On-Screen Switches – Enabled	Shows the virtual on-screen switch panel (see below)
On-Screen Switches – 4 / 8 / 12 / 16	Number of section switches shown on the on-screen switch panel

On-Screen Switch Panel

When enabled, a floating switch panel appears on screen. It stays visible and its buttons track the connected switchbox's live state, but every on-screen button (Master, sections, Auto Rate/Section, Rate Up/Down) becomes unresponsive while a physical switchbox is actively connected — the panel is only functional when no physical switchbox is present. **PRM** is the exception: it always activates Primed Start immediately, even while a physical switchbox is connected.

Button	Function
MST	Toggles master on/off. Green = on, red = off.
PRM	Activates Primed Start immediately. Green while priming is active. Disabled when Maintained switch type is selected or when the vehicle is moving.
↑ / ↓	Rate adjustment. Hold to increase or decrease rate or PWM.
1 – 16	Section switches (4, 8, 12, or 16 buttons, as configured). Green = on, grey = off.
Auto Rate	Toggles automatic rate control.
Auto Section	Toggles automatic section control.

Note: Before pressing PRM, turn on the section switches you want to prime. Only sections that are currently switched on will run during priming. Partial priming (some sections only) is valid.

Controls

Control	Action
 Cancel	Discard changes
 Save	Save settings

15. Machine – Pressure

Menu: Machine > Pressure

Calibrates a pressure sensor using two known reference points, and sets a maximum pressure limit per module.

Calibration Procedure

1. Select the **Module** the pressure sensor is connected to.
2. Set the system to a **known low pressure**.
3. Enter the sensor **reading** and the actual **pressure** value.
4. Set the system to a **known high pressure**.
5. Enter the sensor **reading** and the actual **pressure** value.
6. Enter the **minimum sensor reading** that corresponds to 0 pressure.
7. Press **Show** to display the live pressure screen.

Max Pressure

Sets the maximum allowed pressure for the selected module. When the measured pressure exceeds this value:

- The application shows a **High Pressure** alarm.
- The limit is also sent to the module, which reduces flow on its own to bring the pressure back below the limit — this protects the system even if communication with the PC is lost. Requires up-to-date module firmware.

If the pressure limit trips repeatedly in a short time, the module treats it as a fault and keeps reducing flow until the **master switch is turned off**, which resets the protection.

Set Max Pressure to **0** to disable the limit.

Controls

Control	Action
 Cancel	Discard changes
 Save	Save settings

16. Machine – Primed Start

Menu: Machine > Primed Start

Allows boom priming or a simulated product application while the tractor is stopped. The tractor must not be moving for primed start to activate.

Setting	Description
On Time	How long the application runs (seconds)
Speed	Simulated ground speed used to calculate application rate during priming
Master Switch Delay	How long to hold the physical master switch in the On position to trigger primed start (seconds). Applies to momentary switchbox only.
Resume	After On Time expires, if the tractor is moving, continue with normal operation

Activating Primed Start

Physical Switchbox — Momentary Switch

Hold the master switch in the **On** position for the duration of the **Master Switch Delay**. Primed start activates automatically once the delay expires.

On-Screen Switch Panel

Press the **PRM** button. Primed start activates immediately — no hold required. The PRM button turns green while priming is active.

Before pressing PRM, turn on the section switches you want to prime. Only sections that are currently switched on will run during priming.

Cancelling

Press the master switch Off, or press the on-screen **MST** button while master is on. Primed start stops immediately.

Note: Primed Start is not available when **Maintained** switch type is selected in Machine > Switches.

Controls

Control	Action
 Cancel	Discard changes
 Save	Save settings

17. Modules – Boards

Menu: Modules > Boards

Loads a starting pin layout for a known board type.

Example Presets

These load a starting pin layout for a known board. After loading, edit the Pins, Relay Pins, and Valves pages to match your own board.

Preset	Microcontroller
Load Nano (RC12-3)	Arduino Nano
Load Teensy (RC11-2)	Teensy 4.1
Load ESP32 (RC15)	ESP32

Usage

1. Select the preset that most closely matches your module hardware.
2. Press **Save** to apply the default settings. The Save button highlights when a change is pending.
3. Review and adjust settings on the Config, Pins, Relay Pins, and Valves pages.

Note: Loading a preset overwrites the current module pin and relay settings with defaults. Only do this during initial setup or when switching hardware. The module ID is not changed — a preset always applies to the module selected on the Config page.

Note: The board description and module ID are set on the **Modules > Config** page (see Module Identity, §19).

18. Modules – Communication

Menu: Modules > Communication

Configures how the application communicates with rate control modules. Select either **Ethernet** or **CAN Bus** as the communication type — the two are mutually exclusive. The settings for each are on the **Ethernet** and **CanBus** tabs.

Option	Description
Ethernet	Modules communicate via UDP over the local network. Default mode.
CAN Bus	Modules communicate via a CAN adapter connected to this PC. Ethernet remains active for AgOpenGPS.

Ethernet

Select the PC network interface used to communicate with modules.

Control	Description
IP Address dropdown	Lists active network interfaces on this PC. Select the one on the same subnet as the modules.
Module Subnet	Shows the current subnet the application is using to reach modules.
Send Subnet	Sends the selected IP subnet to the module so it knows which address to respond to.
Rescan	Refreshes the IP address list and module subnet display.

CAN Adapter

Driver	Description
SLCAN	Serial port-based CAN adapter. Requires COM port selection.
InnoMaker USB2CAN	Native USB CAN adapter (InnoMaker hardware). No COM port required.
PCAN	Native USB CAN adapter (Peak PCAN hardware). No COM port required.

COM Port (SLCAN only)

1. Use the dropdown to select the COM port the SLCAN adapter is connected to.
2. Press **Refresh** to rescan available COM ports.

Diagnostics

Option	Description
Diagnostics	Enables CAN traffic debug logging in the activity log

Status Indicators

Indicator	Meaning
Connected	Green — module data received within the last 2 seconds
Driver Found	Green — CAN adapter is open and receiving frames within the last 4 seconds

Note: Changing the CAN adapter type or COM port while CAN Bus is active will restart the CAN bridge with the new settings.

19. Modules – Config

Menu: Modules > Config

Configures the hardware settings of the connected module.

Module Identity

The underlined line at the top shows which module the module pages are editing — the module number, the board's description (or reported label), and its connection state. It updates live while the page is open. If it turns red with **"Two boards are answering as module N!"**, two powered boards share the same module ID — disconnect one and give it its own ID before continuing.

Setting	Description
Module ID	The module to edit. All module pages (Config, Communication, Pins, Relay Pins, Valves) and the Send button work on this module; other connected modules ignore the settings.
New Module ID	Leave blank unless changing the board's module ID. Enter the new ID and press Save; the next Send assigns it to the board. Because every board that hears an ID assignment takes the new ID, only one board may be connected — Send refuses otherwise. After the board restarts and reconnects, press Send once more so the description and sensor pins reach it.
New Board Description	Leave blank to keep the current description. Up to 16 characters (e.g. "Left Tank"), sent to the module with Send and stored in the module's own memory, where it is kept through firmware updates.

Note: A board with no description reports its built-in hardware (MAC) number instead, e.g. **MAC 04E9E5A1B2C3** — unique to every board. This is how RateController tells new boards apart and detects duplicate module IDs. Give each board a friendly description once it is set up. (Nano boards do not report a label.)

1. Relay Type

Value	Description
0	No relays
1	GPIOs — use micro-controller pins directly
2	PCA9555 — 8-relay module
3	PCA9555 — 16-relay module
4	MCP23017
5	PCA9685
6	PCF8574

2. Pin Assignments by Board

Parameter	RC5 (Nano)	RC8 (Nano)	RC11 (Teensy)	RC11-2 (Teensy)	RC12 (Nano)	RC12-3 (Nano)	RC15 (ESP32)
Sensor Count	2	2	2	2	1	2	2
Relay Control	4	4	1	1	2	4	5
Flow 1	2	2	28	28	D3	3	17
Flow 2	3	3	29	29	—	2	16
DIR 1	4	4	37	37	D6	4	32
DIR 2	6	6	14	14	—	6	25
PWM 1	5	5	36	36	D9	5	33
PWM 2	9	9	15	15	—	9	26
Work Pin	—	—	—	30	—	15	—
Pressure Pin	—	—	—	40	—	14	—

3. Nano Pin Reference

Pin Name	Number
D0–D13	0–13
A0–A7	14–21

Controls

Control	Action
 Cancel	Discard changes
 Save	Save settings

20. Modules – Pins

Menu: Modules > Pins

Manually assigns the GPIO pins used for sensors and motor/valve control on the module. The grid shows one row per sensor (the sensor count is set in Modules > Config). Pin values range from 0–50. A blank value (displayed as "–") means the pin is not assigned.

Grid Column	Description
Sensor	Sensor/product number (read-only)
Flow	Input pin for the flow sensor
Dir	Output pin for the motor/valve direction channel
PWM	PWM output pin for the motor/valve
Bin	Input pin for the bin-empty sensor. Leave unassigned ("–") if the product has no bin sensor — the bin alarm is off for that product.
Invert	Inverts the bin sensor reading. Unchecked: an open input reads <i>empty</i> (sensor closes to ground while material is present). Checked: a grounded input reads <i>empty</i> .

Setting	Description
Work Pin	Input pin for the work/implement switch
Momentary	Enables momentary switch behavior on the work pin
Invert Work Switch	Inverts the work switch reading to suit the sensor type. Unchecked: an open input reads as <i>working</i> (normally-closed sensor). Checked: a grounded input reads as <i>working</i> (normally-open sensor). In momentary mode the toggle occurs on the opposite edge. Older module firmware ignores this setting.
Pressure Pin	Analog input pin for the pressure sensor

Press **Rescan** to clear all pin assignments. Duplicate pin numbers are rejected on Save.

Tip: When a board type is selected in Modules > Config, pressing **Defaults** populates these pin values automatically. Only change them if your hardware differs from the standard layout.

Bin-Empty Alarm

When a Bin pin is assigned, the module watches that input and reports the bin state to RateController. The bin sensor input uses an internal pull-up, so a simple switch or level sensor wired between the pin and ground is all that is required. The module filters the reading — the sensor must hold its state for about 2.5 seconds before the reported state changes, so brief flickers from material sloshing are ignored.

- The alarm is only raised **while the master switch is on** — an empty bin at start-up does not alarm until you begin applying.
- When it triggers, the alarm button shows **Bin Empty** with the product name, the product's tile turns red, and on the selected product the quantity caption changes to **Bin Empty** with the quantity value shown as a red 0.

- The displayed 0 is display-only — the recorded tank amount and quantity applied are not changed, and the real value returns as soon as the alarm clears.
- The alarm clears immediately when the bin refills or the master switch is turned off. Tap the alarm button to silence the buzzer.

Tip: Use a plain digital pin for the bin sensor. Inputs with analog conditioning on the board (such as the pressure input) load the internal pull-up and may not read reliably as a digital input.

Controls

Control	Action
 Cancel	Discard changes
 Save	Save settings

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21. Modules – Relay Pins

Menu: Modules > Relay Pins

Assigns the physical GPIO pins to each of the 16 relay outputs on the module. Pin values range from 0–50. A value of 255 (displayed as "–") means the relay pin is not assigned.

- **Low Relays (1–8):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.
- **High Relays (9–16):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.

Press **Rescan** to clear all relay pin assignments.

Tip: When a board type is selected and **Defaults** is pressed in Modules > Config, relay pins are populated automatically for the chosen hardware.

Controls

Control	Action
 Cancel	Discard changes
 Save	Save settings

22. Modules – Valves

Menu: Modules > Valves

Selects the wiring mode for the valve or motor driver.

Option	Description
2-Wire	Motorized or latching valve — the driver actively powers the valve in both directions to open and close it.
3-Wire	One switched signal wire, one power wire and one ground wire. Compatible with standard solenoid valves and motorized ball valves that manage open/close internally. Default for most installations.

Select the mode that matches how your valve or motor driver is wired to the module.

Controls

Control	Action
 Cancel	Discard changes
 Save	Save settings

23. Rate Map

Menu: File > Rate Map

The Rate Map enables variable rate application by defining geographic zones with different target rates. It integrates with AgOpenGPS for GPS-referenced mapping.

Prescriptions Tab

Manages prescription shapefiles for the current field. The active prescription determines which zone rates are used during application.

Control	Action
Field	Shows the field name assigned to the current job
Prescription list	Lists all prescription files available for the current field. Select one to make it active. Zones.shp is the default and cannot be deleted.
New	Create a new empty prescription file with a given name
Copy	Copy the selected prescription to a new name. The copy becomes the active prescription.
Import	Open the prescription import dialog — imports a shapefile or an XML prescription. See Importing a Prescription below.
Export	Export the selected prescription shapefile to a chosen location
Delete	Delete the selected prescription file. Zones.shp cannot be deleted.

Note: Selecting a prescription from the list immediately makes it the active prescription for the current job. The map reloads to show the selected zones.

Importing a Prescription

The Import button opens a two-step import dialog:

Step 1 — Import Data

Control	Description
Shape file / XML	Choose the source format: a prescription shapefile (.shp) or an XML prescription file
Import	Select the file and read it in
Mapping grid	For shapefiles, match each product (A–E) to the attribute column in the shapefile that holds its rate. For XML files, the layer names found in the file are shown.
Default column	Fallback rate per product (shapefile import only). Zones missing the mapped attribute receive this rate instead of 0. Tap the cell to enter a value; leave it at 0 for no fallback.
Product	The primary product (A–E) used when adjusting zones in Step 2. Selected automatically from the imported data; change it if needed.
Name	File name for the saved prescription

Step 2 — Adjust Zones (optional)

Grid-style prescriptions can contain thousands of tiny zones. Adjust consolidates them into fewer, larger zones that are easier to work with and apply.

Setting	Description
Max Zones	Maximum number of distinct rate zones to create
Min zone size	Zones smaller than this area (Ac/Ha) are merged into their neighbours
Min zone step	Minimum rate difference between zones — rates closer together than this are combined
Adjust	Run the consolidation and report how many zones were created

Press **Save** to save the prescription for the current field. It becomes the active prescription. If the imported data has different product names, RC offers to update the job's product names to match.

Zones Tab

Create and edit rate zones within the active prescription.

Create zones by clicking the  **New** button, then clicking at least **3 points** on the map to define the boundary.

- With **overlapping zones**, the **last zone added** takes priority.
- When variable rate is enabled but **no zone exists**, the **base rate** is used.
- Switching to another tab ends an active **New/Edit** — unsaved zone changes are discarded.

Button	Action
 New	Create a new rate zone
 Edit	Edit zone properties (name, rate per product, color)
 Delete	Delete corner point (while creating) or delete zone (while editing)
 Cancel	Cancel current edit
 Save	Save current edit
 Zone List	Open the Zone List — view all zones with their areas, product rates (A–E), and colors; tap a rate cell to change it; select/deselect zones; delete zones; print the list. To set one rate on many zones at once, check the wanted zones, choose the product (A–E), enter the rate and press Apply — then Save or Cancel as usual.

Data Tab

Control	Action
Record count	Number of applied rate data points recorded
A / B / C / D / E	Select which product's applied rate to display on the map
 Record	Start/stop recording applied rates
 Delete	Delete all recorded applied rate data

VR Tab — Variable Rate Look-Ahead

Configures how far ahead of a zone boundary the application rate begins adjusting, so the rate matches the target as soon as the machine enters the new zone. Compensates for applicator response lag.

Setting	Description
Look-Ahead Time (per product)	Seconds before entering a zone boundary that rate adjustment begins. One value per product (up to 5).
Auto Tune	RC monitors look-ahead performance and automatically adjusts the value to improve zone entry accuracy.

Files Tab

Import and manage supporting data files for the current field. Use the file selector dropdowns to choose which file is active for each overlay layer.

Field Boundary

The field boundary defines the operated area and is used to clip automatically generated productivity zones. Select the active boundary from the **KML** dropdown. Import a boundary using the **Import Boundary** button — three file types are accepted:

- **AOG Field.kml** — the field KML from the AgOpenGPS Fields folder; the boundary polygon is extracted automatically.
- **AOG Boundary.txt** — AgOpenGPS boundary file; easting/northing coordinates are converted to lat/lon using the field origin.
- **Any KML (*.kml)** — any KML file containing a polygon boundary.

Tip: Field files are typically found in:

- **AgOpenGPS:** Documents\AgOpenGPS\Fields\[Field Name]\ — contains `Field.kml` and `Boundary.txt`
- **TWOL:** Documents\TWOL\Fields\[Field Name]\ — contains `Boundary.txt`

Yield, EC, and Elevation Data

Control	Action
Yield dropdown	Select the active yield data file for the overlay and zone creation
Import Yield	Import a yield data CSV file. If the file contains elevation data, RC will offer to extract it to a separate elevation file.
Delete Yield	Remove the selected yield data file
EC dropdown	Select the active EC (electrical conductivity) data file
Import EC	Import an EC data CSV file (Lat, Lon, EC)
Delete EC	Remove the selected EC data file
Elevation dropdown	Select the active elevation data file
Import Elevation	Import an elevation data CSV file (Lat, Lon, Elevation)
Restore Elevation	Restore the previous elevation file from the automatic backup
Delete Elevation	Remove the current elevation data file

Create Zones

The **Create Zones** button opens the productivity zone creation dialog. Zones are generated automatically from yield history, soil EC, elevation data, or any combination. Clear existing zones before generating if you want to replace them.

Setting	Description
Yield files (checklist)	Check one or more yield data files to include. Multiple years are blended by their assigned weight.
EC	Include the active EC layer in zone generation
Elevation	Include the active elevation layer in zone generation
Weights	Percentage weight for each active layer. All weights must sum to 100 before Build is enabled.
Productivity levels	Number of productivity classes to create (e.g. 3 = Low / Med / High). Each class may produce more than one zone polygon if the field has non-contiguous areas of the same productivity level.
Min zone size	Minimum zone area in hectares (or acres). Zone polygons below this size are discarded. Increase to reduce the number of small fragments.
Build	Generate zones using the selected layers and settings. The field boundary (set in the KML dropdown) clips zones to the operated area.

Tip: Set the field boundary KML before generating zones. Without a boundary, zones are clipped to the bounding box of the data files, which may not match the actual field edge.

Display Toggle Buttons (right panel)

These buttons highlight green when active. Click to toggle on or off.

Button	Action
Sat. View	Toggle satellite background image
Target Zones	Toggle rate zone boundaries display
Applied Rates	Toggle applied rate colour overlay
KML Files	Toggle KML file overlay
Yield Data	Toggle yield data colour overlay
EC	Toggle soil EC colour overlay
Elev.	Toggle elevation colour overlay

Bottom Controls

Button	Action
 Minimize	Shrink the map to a small window. Drag it by its title bar to reposition.
 Centre	Centre the map on the current GPS position
 Help	Open the help page
 OK	Close the Rate Map

24. Rate Graph

Menu: **Help > Rate Graph** (opens as a separate window)

Displays a real-time chart of the PID control loop for a product. Used for tuning Gain and Integral settings.

Displayed Data

Element	Description
Target UPM line	The desired flow rate setpoint
Applied UPM line	The measured actual flow rate
Error %	Current deviation from target
PWM value	Current PWM output (0–255)
Frequency (Hz)	Current flow sensor frequency

Chart Controls

Control	Description
Auto	Auto-scale the Y-axis to fit the data
Up	Increase Y-axis range by 1.5×
Down	Decrease Y-axis range by 0.67×

The chart maintains the last 30 data points. Click the graph to select which product to display.

25. Scale / Tank Display

Menu: Products > Settings > Scale Weight (enable checkbox to show)

Displays a floating window showing tank/scale data for a product. Each product has its own scale display window.

Display Modes

Left-click on the displayed value to cycle through four modes:

Mode	Description
Current Weight	Net weight from the scale sensor (after tare)
Applied Weight	Difference between starting weight and current weight
Area Covered	Area worked since the last reset
Application Rate	Applied weight divided by area covered

Controls

Action	Result
Left-click on value	Resets the starting weight and acres reference point
Right-click on value	Resets the tare (zero calibration)
Right-click and drag	Repositions the window

26. Language Selection

Menu: File > Language

Changes the display language of the application.

Supported Languages

- English
- Deutsch (German)
- Hungarian
- Nederlands (Dutch)
- Polish
- Russian
- French
- Lithuanian

Note: A restart of the application is required for the language change to take effect.

27. Color Customization

Menu: File > Color

Customizes the foreground and background colors of the application interface.

Preset Color Schemes

Three built-in color presets are available. Select one to apply it immediately.

Custom Color

Select **Custom Color** to open the color picker:

1. Click in the **color gradient panel** to select a hue and saturation.
2. The horizontal position controls **hue**.
3. The vertical position controls **brightness/saturation**.
4. Separate foreground and background colors can be selected.

Color preferences are saved and applied immediately across all screens.

28. Help – Logs & Diagnostics

Menu: Help (main menu)

Provides access to application logs, module diagnostics, and system information.

Log Files

Log	Description
 Activity Log	Records application events and UDP/CAN communication activity in real time
 Error Log	Records application errors and exceptions
 Ethernet Log	Records raw UDP packet traffic — useful for diagnosing communication issues

Press **Start/Pause** to freeze or resume live log updates.

Activity Log — Module Status Events

When a module connects or its status changes, RC logs the following to the Activity Log automatically:

Log Entry	Description
Module X, Ethernet connected: True/False	Whether the module's Ethernet port is connected
Module X, Wifi Strength: 0–3	ESP32 WiFi signal level — 0 = not connected, 1 = weak (RSSI < -80), 2 = moderate (RSSI < -70), 3 = good (RSSI < -65)
Module X, Valves are 2 Wire / 3 Wire	Valve wiring mode reported by the module
Module X, Pin Configuration correct / not correct	Whether the module's pin configuration is valid

These entries are written once on first connect and again whenever the status changes.

Controls

Control	Action
 Start/Pause log	Freeze or resume live log updates
 Reset logs	Delete all log files after confirmation
PID Logging	Toggle PID diagnostic logging. While on (green), the module streams per-loop rate control data (target, applied rate, PWM, error) which is saved to a CSV file in the <code>PIDLogs</code> folder inside the data folder — useful for tuning and diagnosing rate control problems. The setting is remembered between sessions.
 Open rate graph	Display the rate graph
 Check for updates	Check for a newer version of the application
 Open user manual	Open the user manual
 Navigate modules left	View previous module diagnostics
 Navigate modules right	View next module diagnostics

Module Diagnostics

Display	Description
Module ID	ID of the module currently being viewed (use left/right arrows to switch)
Firmware Version	Firmware version running on the selected module
Elapsed Time	Seconds since last module data was received

System Information

Display	Description
IP / Subnet	Current Ethernet subnet in use
Application Version	Installed RC software version and date
Current File	Name of the active profile file
GPS Coordinates	Live latitude and longitude from AgOpenGPS

External Links

Links to the project repositories are available from this screen:

- [AOG RC GitHub repository](#)
- [PCB Setup GitHub](#)
- [Rate Control PCBs GitHub](#)

29. Tips & Tricks

Main Screen Navigation

- **To reposition the main screen**, press and hold (or right-click) in the middle of the screen, then drag it to the desired location.
- Use the **minimize button** on the main form to reduce the display to rate only. RC remembers this — if it was minimized when closed, it starts minimized next time.

Clickable Display Elements

- **Click on target rate label** to switch between the target rate and the alternate rate (a percentage of the target). When variable rate is enabled, the label shows the VR target rate and is not clickable.
- **Click on quantity applied label** to switch between quantity applied and quantity remaining.
- **Click on area applied label** to switch between area applied and area remaining.
- **Click on area applied value** to reset the area counter.
- **Click on the tank bar graph** to reset the tank quantity.
- Click a **product icon** at the top of the screen to select which product's rates and totals are displayed. The rate graph will reflect the currently selected product.

30. Troubleshooting

No Relays Activating

1. Is the module Config set correctly for board type and relay type?
2. Has the subnet been set in the app and uploaded to the module?
3. Have section count and widths been set?
4. Is Auto mode on?
5. Is a section assigned to each relay?
6. Is a switch assigned to each section?
7. If Auto is on, have sensor counts/unit and base rate been set?

No UPM (No Flow Reading)

1. Check ground speed.
2. Check working width.
3. Check sensor control settings.
4. Is AgOpenGPS connected?
5. Is the module connected?
6. Check sensor counts/unit.
7. Check base rate.
8. Have control values been sent to the module?
9. Check control type — is it a valve or motor?
10. Is the connector fully plugged in?
11. Is the module running the latest firmware?
12. Is Auto mode on?

AOG Status Light Not Green

1. The AOG light reflects whether AgOpenGPS is sending GPS or AutoSteer position data over the network — RateController does not need to send anything for it to turn green.
2. AgOpenGPS only sends that data once it has an active position source: a real GPS fix, or its built-in GPS Simulator turned on. No fix and no simulator means the light stays red, even if RateController and AgOpenGPS are both running correctly.
3. If your AutoSteer hardware also supplies GPS (common on combined GPS/IMU/steer boards), unplugging it removes the position source too — the AOG light will drop even though AutoSteer itself isn't what RateController is waiting on.
4. To test the connection without field hardware, turn on AgOpenGPS's GPS Simulator — the light should turn green the same way it would with a real fix.

No Rate Adjustment

1. In manual mode, the master switch must be on.
2. Is minimum PWM too low?

Motor Won't Shut Off

1. Minimum PWM should be set to 0.
2. Is master override enabled?

After Updating Module Firmware

1. Resend the subnet (Modules > Communication).
2. Resend all module config settings (Modules > Config, then Send).

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